

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch: Electrical & allied

Semester : VII

Subject Code & Name: Power System Operation and Control (BTEEC702)

Max Marks: 60

Date:04-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	Level/CO	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain production and absorption of reactive power by different power system equipment's.	L2	6
B) Enlist the methods of voltage control and explain any one of them.	L2	6
C) Explain the construction and working of Static VAR Compensator. List down its advantages.	L2	6
Q.2 Solve Any Two of the following.		12
A) Derive swing equation for a synchronous generator.	L3	6
B) Explain the equal area criteria in detail with any one application.	L3	6
C) State and explain in short different methods to improve steady state and transient stability.	L2	6
Q. 3 Solve Any Two of the following.		12
A) Explain the construction and working of DC excitation system with neat sketch.	L2	6
B) What control and protective functions does an excitation system perform? Explain with block diagram.	L2	6
C) What are different elements of an excitation system? Explain with the help of block diagram.	L2	6
Q.4 Solve Any Two of the following.		12
A) Explain the construction and working of Speed governing mechanism with neat diagram.	L2	6
B) What is speed governor dead band? Explain its effect on automatic generation control.	L2	6
C) The B matrix for a two plant system is given by:	L3	6

$$B = \begin{bmatrix} 0.001 & -0.0001 \\ -0.0001 & 0.0013 \end{bmatrix} MW^{-1}$$

The economic schedule for a certain load is given as: P1 = 150 MW and P2 =

275 MW. Calculate the penalty factor of plant 1 and plant 2

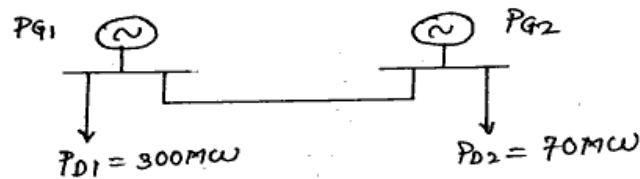
Q. 5 Solve Any Two of the following.

12

- A) A two bus power system is shown in Fig.

L4

6



Incremental fuel costs of two generators are given as

$$IC_1 = 0.31P_{G1} + 41 \text{ Rs/MWhr}$$

$$IC_2 = 0.35P_{G2} + 41 \text{ Rs/MWhr}$$

Loss Expression as $P_L = 0.001(P_{G2} - 70)^2 \text{ MW}$

Determine optimal scheduling and power loss of transmission link.

- B) Explain the constraints with respect to Unit Commitment.

L2

6

- C) The fuel input per hour of plant 1 and plant 2 is given as:

L3

6

$$F_1 = 0.2P_1^2 + 40P_1 + 120 \text{ rupees per hour}$$

$$F_2 = 0.25P_2^2 + 30P_2 + 150 \text{ rupees per hour}$$

Determine the economic operating schedule and the corresponding cost of generation if the minimum and maximum loading on each unit is 25MW and 100 MW respectively. Total demand is 180 MW and transmission losses are neglected. If the load is equally shared by both the units, determine the saving obtained by loading the units as per equal incremental fuel cost.

***** All The Best *****